CLOUD COMPUTING PRACTICALS

Short Programs / Commands with the Same Sample Outputs

Prepared from the uploaded practical record. The programs and commands are shortened for easier writing; sample outputs follow the record. Replace example resource names, IP addresses, project IDs, and IAM role ARNs with the values from your own cloud account when performing the experiments.

EXPERIMENT 1 — Google Drive Spreadsheet and Notes

AIM: To create, edit and share a spreadsheet and a note/document using Google Drive.

PROGRAM / COMMANDS:

```
Google Sheets: enter the following rows and use =SUM(C2:E2) in F2, then fill down.

Roll No | Name  | Mark1 | Mark2 | Mark3 | Total
101     | Arun  | 85    | 90    | 88    | =SUM(C2:E2)
102     | Priya | 78    | 82    | 91    | =SUM(C3:E3)
103     | Kumar | 92    | 88    | 85    | =SUM(C4:E4)

Create a Google Docs note, rename it Cloud_Notes, and share with Editor access.
```

SAMPLE OUTPUT:

```
Roll No  Name   Mark1  Mark2  Mark3  Total
101      Arun   85     90     88     263
102      Priya  78     82     91     251
103      Kumar  92     88     85     265

Document "Cloud_Notes.docx" created and shared successfully with Editor access.
```

RESULT: Thus a spreadsheet and a note document were created, edited and shared using Google Drive.

EXPERIMENT 2 — Launch a Linux Virtual Machine

AIM: To launch a Linux virtual machine on Amazon EC2 and connect remotely.

PROGRAM / COMMANDS:

```
chmod 400 CloudKey.pem
ssh -i "CloudKey.pem" ubuntu@<Public-IP-Address>
uname -a
df -h
```

SAMPLE OUTPUT:

```
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-1019-aws x86_64)
Linux ip-172-31-40-10 5.15.0-1019-aws #23-Ubuntu SMP x86_64 GNU/Linux

Filesystem   Size  Used  Avail  Use%  Mounted on
/dev/xvda1   7.6G  1.6G  6.0G   21%   /
```

RESULT: Thus a Linux virtual machine was launched on AWS EC2 and accessed remotely using SSH.

EXPERIMENT 3 — Host a Static Website

AIM: To host a static website using an Amazon S3 bucket.

PROGRAM / COMMANDS:

```
<!-- index.html -->
<!DOCTYPE html>
<html>
<head><title>My Cloud Website</title></head>
<body>
  <h1>Welcome to My Static Website</h1>
  <p>Hosted on Amazon S3.</p>
</body>
</html>

Enable Static website hosting in S3 and set index.html as the index document. Grant public s3:GetObject access using the
```

SAMPLE OUTPUT:

```
Browser displays:
Welcome to My Static Website
Hosted on Amazon S3.
```

RESULT: Thus a static website was hosted using Amazon S3.

EXPERIMENT 4 — Explore Google Cloud Services

AIM: To explore services offered by Google Cloud Platform.

PROGRAM / COMMANDS:

```
gcloud projects list
gcloud compute instances list
gsutil ls
bq query --use_legacy_sql=false \
  'SELECT name, count FROM `bigquery-public-data.usa_names.usa_1910_2013` LIMIT 5'
```

SAMPLE OUTPUT:

```
NAME      ZONE           MACHINE_TYPE  STATUS
demo-vm   us-central1-a  e2-micro      RUNNING
```

RESULT: Thus Google Cloud Platform services were explored successfully.

EXPERIMENT 5 — Google App Engine

AIM: To configure Google App Engine and deploy a sample application.

PROGRAM / COMMANDS:

```
# main.py
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return 'Hello, World! This is my first App Engine application.'

# app.yaml
runtime: python39

# Deployment commands
gcloud init
gcloud app deploy
gcloud app browse
```

SAMPLE OUTPUT:

```
Services to deploy:
descriptor: [app.yaml]
source: [/home/user/myapp]
target project: [cloud-lab-project]
target service: [default]
Deployed service [default] to [https://cloud-lab-project.uc.r.appspot.com]

Browser output: Hello, World! This is my first App Engine application.
```

RESULT: Thus a sample application was deployed successfully using Google App Engine.

EXPERIMENT 6 — Microsoft Azure

AIM: To create and manage resources using Azure CLI.

PROGRAM / COMMANDS:

```
az login
az group create --name CloudLabRG --location eastus
az vm create --resource-group CloudLabRG --name CloudLabVM \
  --image UbuntuLTS --admin-username azureuser --generate-ssh-keys
az resource list --resource-group CloudLabRG --output table
```

SAMPLE OUTPUT:

```
Name         ResourceGroup  Location  Type
CloudLabVM   CloudLabRG     eastus    Microsoft.Compute/virtualMachines
CloudLabVMNic CloudLabRG     eastus    Microsoft.Network/networkInterfaces
CloudLabVMDisk CloudLabRG    eastus    Microsoft.Compute/disks
```

RESULT: Thus Azure CLI was used to create and manage cloud resources.

EXPERIMENT 7 — Connect Redshift with S3

AIM: To load data from an S3 bucket into an Amazon Redshift table.

PROGRAM / COMMANDS:

```
CREATE TABLE employee (
  emp_id INT,
  emp_name VARCHAR(50),
  dept VARCHAR(50),
  salary DECIMAL(10,2)
);

COPY employee
FROM 's3://cloudlab-bucket/employee.csv'
IAM_ROLE 'arn:aws:iam::123456789012:role/RedshiftS3Role'
CSV IGNOREHEADER 1;

SELECT * FROM employee;
```

SAMPLE OUTPUT:

```
emp_id | emp_name | dept    | salary
1      | Arun     | IT      | 45000.00
2      | Priya    | HR      | 38000.00
3      | Kumar    | Finance | 52000.00
```

RESULT: Thus Amazon Redshift was connected with S3 and the data was loaded and queried.

EXPERIMENT 8 — Create and Query a NoSQL Table

AIM: To create and query a NoSQL table using Amazon DynamoDB.

PROGRAM / COMMANDS:

```
aws dynamodb create-table \
  --table-name Student \
  --attribute-definitions AttributeName=StudentID,AttributeType=S \
  --key-schema AttributeName=StudentID,KeyType=HASH \
  --billing-mode PAY_PER_REQUEST

aws dynamodb put-item --table-name Student \
  --item '{"StudentID":{"S":"101"},"Name":{"S":"Arun"},"Marks":{"N":"85"}}'
```

SAMPLE OUTPUT:

```
{
  "Item": {
    "StudentID": {"S": "101"},
    "Name": {"S": "Arun"},
    "Marks": {"N": "85"}
  }
}
```

RESULT: Thus a NoSQL table was created and queried using Amazon DynamoDB.